

CLOCKS

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The dial of a clock is a circle which is divided into 60 min spaces.

The min hand passes over 60 min spaces while the hour hand goes over 5 min spaces.

So in 60 min, the min hand gains 55 min on the hour hand.

Gain of 55 min when actual movement = 60 min

Gain of 1 min when actual movement = $60/55$ min = $12/11$ min

Minute hand gains 1 min over hours hand for every $12/11$ min of actual movement.

KEY POINTS :

- 1 min space = $360^\circ/60 = 6^\circ$ (As 360° of the circle is divided into 60 min)
- In 1 min, the min hand moves 6° angle.
- In 1 min, the hour hand moves =
 $360^\circ/(12 \times 60) = 360^\circ/720 = 1^\circ/2$
(12 h = 60 min each)
Hence, in 1 min, the min hand gains $5\frac{1}{2}^\circ$ over the hour hand.
- In every hour,
 - (i) The hands coincide once (0° apart)
 - (ii) The hands of clock are at right angle (when the distance between two hands measures 90°) when they are 15 min spaces apart. This situation occurs twice in every hour.
 - (iii) The hands of a clock are opposite (180°) to each other when they are 30 min spaces apart. This situation occurs once in every hour.
- The hands of the clock are in the same straight line when they are coincident or opposite to each other.
- The hands coincide 11 times in every 12 h (between 11 and 1 o'clock there is a common position at 12 o'clock).
Hence, the hands coincide 22 times in a day.
- If both the hands start moving together from the same position, both the hands will coincide after $(360 \times 2)/11 = 65\frac{5}{11}$ minutes.
- The hands of a clock are at right angles twice in every hour, but in 12 h they are at right angle 22 times. Since two positions (3 o'clock and 9 o'clock are identical). Hence, the hands are at right angle 44 times in a day.